CSE-321

The Transport Layer

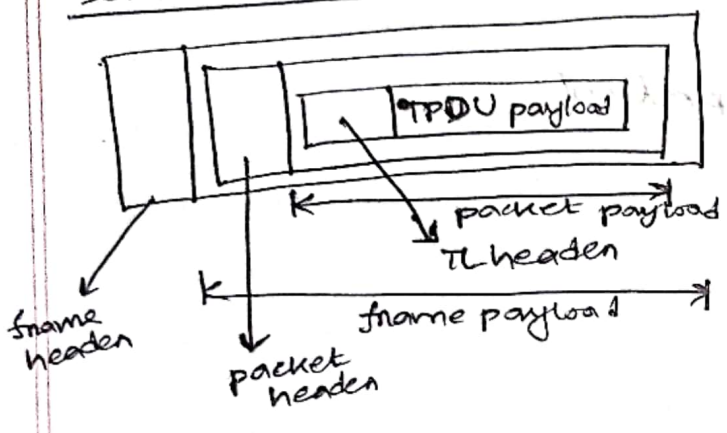
07/01/2026

The Transport Service

↳ uses sockets

↳ TPDU ⇒ transport protocol data unit

service primitives



{ BIND code

DLL → next node → physical comm. channel

TL → end to end → may be 1 hop too

↳ addn: port

Two-army problem

3 way handshake → done

connection establishment

↳ requires handshake

→ conn. req →

← ack ←

→ ack →

⋮ loop

Crash recovery → X

FSM

active → req → no reply → timeout

protocols → UDP, TCP

↓

user datagram protocol

↓

- connectionless
- no error control
- " flow "
- " congestn "

UDP

(16)	(16) bits
src port	destn port
UDP len	UDP checksum
(16)	(16)

used when ack & continuing all pckts is problematic

↳ video, RPC

↳ multimedia

↳ can use TCP too

↳ (2) conditions

↳ book

RPC

client → client stub → pack params (Marshaling)

server stub → unpacking → server

issue → global vars, ptrs, end of str marker (differs), data unit size may vary

(I) (II) (III) (IV)

RTP → above TL → may be application layer

multiple src multimedia stream

Ver

↳ by default 2

padding → 1 → ✓

1 word rows

(4)

CC → contributing src

4 bits

↳ not max 15 srcs

M → marker → varies from appl to appl

payload type

seq# → video frame seq #

timestamp → first frame ~~type~~ time

synchronization src identifier

↳ at least one src

contributing src

TCP → guaranteed tx
duplex
→ transmission control protocol

error
flow
congestn } control

http → 80
https → 443

header

src port
destn port
seq # → for ack, frame
ack # → for ack rcv → what to send if no ack to send

1 byte data + IP → 20B min + TCP → 20B min → 41B

TCP header len → max 15W → 60B
URG → urgent flag → process this immediately
ACK → 1 → ack # is valid

data → queued
send some data to front → PSH → 1

RST → conn. reset → 1
reason { → didn't accept req
→ invalid segment rcvd

FIN → finish the tx → 1
↳ conn. terminate

first req
↳ synchronized req
↓
SYN bit → 1
↓
conn.
~~terminate~~
establish

flow ctrl ⇒ send act rcvr cap.
window size → available at rcvr } sends
urgent ptr → ptr to urgent part
checksum →
↳ if this is corrupted
↳ discard pckt

checksum also includes src IP, dst IP, pad...

breaks all to 16 bits

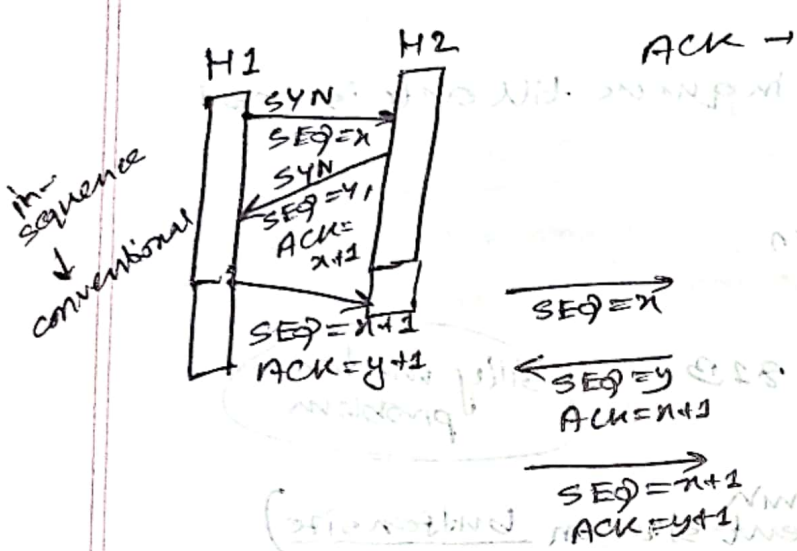
↳ 1's complement of all...

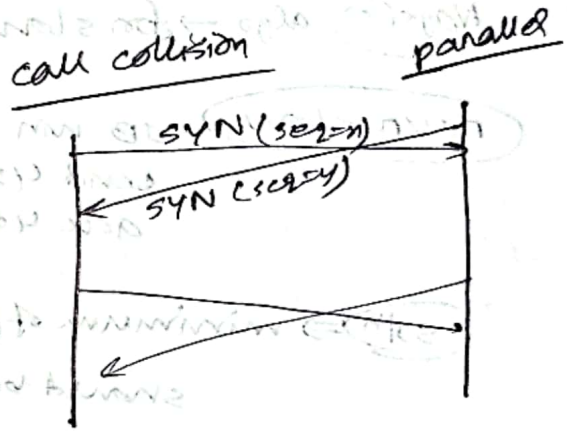
rcvr → check → non-zero after process → discard

ACK → ② conventions

↳ ack the next one

↳ ack the next expected one

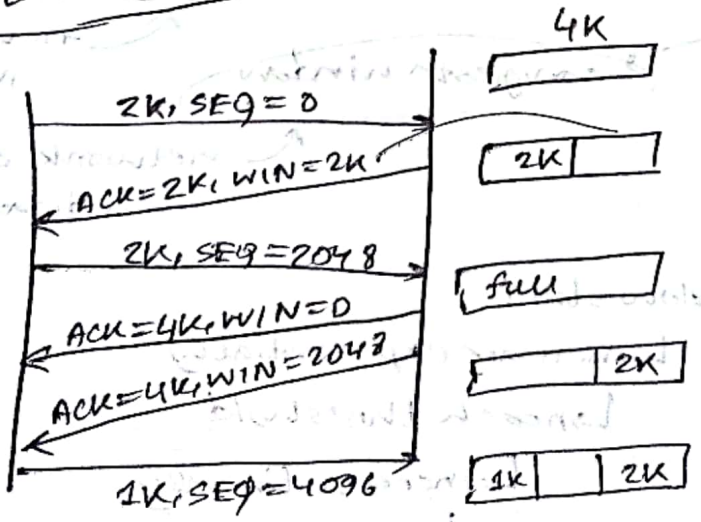




connection mgmt modeling

terminate → FIN/ACK

transmission policy



typing on a remote pc

↳ each byte → 41B pckt 1B → 162B

ack → no data → 40B min highly inefficient

buffer empty → send window size → 40B

TelNet → screen char showed → 41B echo

opt ⇒ send, keep next chars in queue till ack is recvd

5B / 10B → 162B - -

?

Najel's algo → for slow senders

recvr slow ... 1B min
send 41B
ack 40B } 81B silly window problem

soln ⇒ minimum of (min segment size or buffer size / 2)

should be empty → TCP segment

don't advertise otherwise

Clarke's algo

recvr window → for flow ctrl

← recvr cap indicator

congestn ctrl → congestn window

↳ network capacity

← network cap indicator

start with small window

TCP Tahoe

slow start

↳ increase exponentially

↳ reach threshold

↳ increase linearly 39, 192

↳ timeout

↳ window = 1

↳ threshold = timeout cap / 2

timeout

DLL → 1 hop → time ≈ RTT → squeezed dist.

TL → many hops → distribution → sparse

↳ small timeout → valid tx timedout

↳ large " → many time outs occurred already

Wireless TCP & UDP

↳ unreliable, shared

→ timeout → not necessarily cap full

↳ not enough

Tanenbaum

→ next → skip